

Question	Marking Guidance	Mark	Comments
09.1	(Has) phosphate OR (Has) deoxyribose OR (Has) hydrogen bonds;	1 (1 x AO2)	Ignore the number of hydrogen bonds Accept both contain a pyrimidine/single ring (structure) Accept 'H bonds'

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09.2	Correct answer of 8×10^{-6} OR 8.3×10^{-6} = 2 marks ;; Incorrect answer of 0.000 008 3 (correct answer but not in standard form) = 1 mark OR 8.3×10^{-8} (correct division using correct number of G-C pairs, and in standard form, but not shown as a percentage) = 1 mark OR Correct answer in incorrect standard form; eg 83×10^{-7} = 1 mark OR 1.2×10^9 (correct number of G-C pairs in the genome in standard form) = 1 mark ;	2 (2 x AO2)	Accept any number of decimal places that round to 8.3

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09.3	1. Substitution (mutation occurred): 2. (Only) one nucleotide/base pair is changed (in a gene) OR (Only) one (DNA) triplet/codon changed; 3. Same amino acid (coded for); 4. (Because) DNA/genetic code is degenerate; 5. (So) tertiary structure is not changed; 6. (Change) could be in an intron; 7. Removed during splicing;	4 max (4 x AO2)	3. Reject same amino acid is produced 3. Accept one amino acid changed 4. Accept a description of degenerate code 3 and 4 can be awarded together, e.g 'different codons/ triplets code for the same amino acid' = MP3 and MP4

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09.4	1. No (functional) enzyme/ X ; 2. (So) more/faster cell cycles; 3. More(frequent) DNA replication OR DNA replication not delayed; 4. (So) mutations (more likely to) occur in DNA replication;	3 max (3 x AO2)	2. Ignore 'cell cycle isn't slowed down' on its own 3. Accept 'faster DNA replication'